Initial Project Planning Template

Date	18 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	5 Marks

Project timeline: 15 September 2026 to 29 September 2026. Ideation, requirements and design were completed on 15-17 September; development and delivery were organised into three 4-day sprints (18-29 September).

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Project Setup	USN-1	As a developer, I can set up the Node.js/Express project with dotenv, cors, MongoDB connection and the src folder structure.	3	High	S. Danielraj	18 Sep 2026	21 Sep 2026
Sprint-1	Registration	USN-2	As a user, I can register by entering my name, email and password, which is validated and stored with a bcrypt-hashed password.	3	High	Vignesh Kumar	18 Sep 2026	21 Sep 2026
Sprint-1	Login	USN-3	As a user, I can log in with email and password and receive a JWT token.	3	High	Vignesh Kumar	18 Sep 2026	21 Sep 2026
Sprint-1	Profile	USN-4	As a logged-in user, I can view my profile through a JWT-protected route.	2	High	Arul Edvish	18 Sep 2026	21 Sep 2026
Sprint-1	Error Handling	USN-5	As a client, I receive clean, consistent error responses (duplicate email, validation, invalid ID) from a central error handler.	3	High	Roydon Malachi Martin	18 Sep 2026	21 Sep 2026
Sprint-1	Validation	USN-6	As a client, my requests are checked by validation middleware before reaching the controllers.	3	Medium	Roydon Malachi Martin	18 Sep 2026	21 Sep 2026
Sprint-1	Data Models	USN-7	As a developer, I have User and FAQ Mongoose schemas with required fields, category enum and createdBy reference.	3	High	S. Danielraj	18 Sep 2026	21 Sep 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-2	FAQ Management	USN-8	As a creator, I can create an FAQ with question, answer and category.	3	High	Arul Edvish	22 Sep 2026	25 Sep 2026
Sprint-2	FAQ Management	USN-9	As any user, I can list all FAQs and open one FAQ by its ID.	2	High	Arul Edvish	22 Sep 2026	25 Sep 2026
Sprint-2	FAQ Management	USN-10	As a creator, I can update or delete only the FAQs I authored.	3	High	Arul Edvish	22 Sep 2026	25 Sep 2026
Sprint-2	Search	USN-11	As a reader, I can search FAQs by keyword using /api/faqs/search?q=.	3	High	Roydon Malachi Martin	22 Sep 2026	25 Sep 2026
Sprint-2	AI Integration	USN-12	As a developer, I have a Gemini service that wraps the @google/genai SDK.	3	High	S. Danielraj	22 Sep 2026	25 Sep 2026
Sprint-2	AI Integration	USN-13	As a logged-in user, I can ask a question and get an AI-generated answer.	3	High	S. Danielraj	22 Sep 2026	25 Sep 2026
Sprint-2	AI Integration	USN-14	As a creator, I can give a topic and receive an AI-generated FAQ (question, answer, category).	3	High	Vignesh Kumar	22 Sep 2026	25 Sep 2026
Sprint-3	Testing	USN-15	As a team, we test all endpoints in Postman and keep a Postman collection.	5	High	Vignesh Kumar	26 Sep 2026	29 Sep 2026
Sprint-3	Testing	USN-16	As a team, we add a test_api.js script and fix issues found during testing.	3	Medium	Roydon Malachi Martin	26 Sep 2026	29 Sep 2026
Sprint-3	Documentation	USN-17	As a team, we document architecture, ER diagram, user flow and the API testing specification.	5	High	S. Danielraj	26 Sep 2026	29 Sep 2026
Sprint-3	Demo	USN-18	As a team, we prepare and rehearse the project demonstration.	4	Medium	All members	26 Sep 2026	29 Sep 2026
Sprint-3	Performance	USN-19	As a team, we record response times of key endpoints.	3	Medium	Arul Edvish	26 Sep 2026	29 Sep 2026

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	4 Days	18 Sep 2026	21 Sep 2026	20	21 Sep 2026
Sprint-2	20	4 Days	22 Sep 2026	25 Sep 2026	20	25 Sep 2026
Sprint-3	20	4 Days	26 Sep 2026	29 Sep 2026	20	29 Sep 2026

Velocity

Sprint duration is 4 days and the team velocity is 20 story points per sprint. Average velocity (AV) per iteration unit = 20 story points / 4 days = **5 story points per day**. Total scope: 60 story points across 19 user stories.

Burndown Chart

